

Proposal: Biney's Procedural Compression

This paper proposes a new research direction for the development of a compression algorithm for the BLUE system. The idea is currently in its early stages but provides a foundation for further research and experimentation. It is expected to evolve through discussion, testing, and refinement, including the possible use of artificial intelligence and reinforcement learning to improve the search process.

The proposed method, referred to as *Biney's Procedural Compression*, aims to find the smallest possible program capable of reconstructing a file losslessly. Rather than representing a file directly as compressed data, the compressor attempts to discover a compact procedural description of the file.

A procedural description consists of:

- An algorithm.
- A seed (or set of parameters).

Executing the algorithm together with its seed should reproduce the original file exactly.

The objective is therefore to discover, as efficiently as possible, the smallest program capable of reconstructing the original data without loss. If the resulting procedural description is substantially smaller than the original file, it serves as the compressed representation.

This concept is closely related to Kolmogorov complexity, which studies the length of the shortest program capable of generating a given string. Since the exact shortest program is not computable in the general case, the objective is instead to search for practical approximations under realistic computational constraints.

Example constraints include:

- Maximum program size $\leq K$ bytes.
- Reconstruction time $\leq T$.
- Search time or computational budget $\leq C$.

Within these constraints, the compressor searches for the best procedural representation it can find.

Several possible extensions naturally follow from this framework:

- AI or reinforcement learning to guide the search.
- Hybrid approaches combining procedural compression with existing lossless compression algorithms.
- Domain-specific procedural generators for images, audio, video, documents, and 3D assets.

This proposal is intended as a research direction rather than a finalized compression algorithm. The underlying algorithms, optimization methods, and practical constraints remain open for future investigation.

One of the primary motivations for exploring this approach is the long-term archival of the BLUE system. If Biney's Procedural Compression discovers sufficiently compact procedural representations for large portions of the BLUE archive, or even the archive as a whole, those procedural programs could be distributed across a peer-to-peer network as a decentralized archive.

Combined with techniques such as erasure coding or threshold reconstruction, where only K out of P distributed fragments are required to recover the complete archive, this approach could improve the efficiency, resilience, and decentralization of long-term BLUE storage.

Although archival storage represents only one possible application of this research, it is one of the primary motivations for pursuing this direction.